# MPB 小型化金属化聚丙烯膜电容器(盒装型) 105°C

MPB Mini-size Metallized Polypropylene Film Capacitor (Box-type) 105°C




**系列 Series** MPB

**行业型号 Industry code** MKP25

**产品类型 Type**

小型化金属化聚丙烯膜电容器(盒装型)
Mini-size Metallized Polypropylene Film Capacitor (Box-type)

**应用 Applications**

- 用于开关电源，电子镇流器和变频器等中间电路直流滤波(DC-link, PFC 等)
  As intermediate circuit capacitor for SMPS, Electronic Ballast inverter (i.e. DC-link, DC filter and P.F.C)

## 技术要求 Specifications

| 项目 Item | 参数 Specification |
|---|---|
| **引用标准** Reference Standard | GB/T 10190 (IEC 60384-16) |
| **气候类别** Climatic Category | 40/105/21 |
| **工作温度范围** Operating Temperature Range | -40°C~+105°C (+85°C~+105°C): derating factor 1.5% per °C for R.V(DC) |
| **额定电压** Rated voltage | 250VDC, 400VDC/450VDC, 500VDC/550VDC, 630VDC |
| **容量范围** Capacitance range | 0.01μF~2.2μF |
| **容差** Capacitance tolerance | ±5% (J), ±10% (K) |
| **耐电压** Voltage Proof | 1.6* R.V(DC), 2s (between terminals) |
| **损失角** Dissipation Factor | ≤0.1% (1kHz at 20~25°C) |
| **绝缘阻抗** Insulation Resistance | ≥30 000MΩ, $C_N$≤0.33μF; $RC_N$≥10 000 s, $C_N$>0.33μF (20°C, 100V, 1min) |

## 使用注意 Usage Note

注：MPB 产品只适用于 DC 滤波和 DC 模块电路，意指施加给电容的电压必须是单向涟波电压，电压代表波形如下图示。不适用于交流电路，对此有何疑问请和我们的技术人员联系。

a: The MPB series is only recommended to use in DC-filter or DC-blocking circuits. It means the voltage applied to the capacitors must be unidirectional ripple voltage. The typical voltage curve is as following reference. If you have any questions for this note, please feel free to contact our technical engineer.

Here: V1≥0, V2≤$U_R$, Vrms=(V2−V1)÷√2, Irms=2πf×C×(V1−V2)÷√2; $U_R$ is the rated voltage of the capacitor

## 最大脉冲爬升速率 Maximum Pulse Rise Time

### 最大脉冲爬升速率 dV/dt (V/μs) Maximum Pulse Rise Time (dV/dt)

| $U_R$ (V) | P=7.5 | P=10 | P=15 | P=22.5 |
|---|---|---|---|---|
| **400/450** | 320 | 300 | 200 | 100 |
| **500/550** | 380 | 350 | 220 | 150 |
| **630** | 420 | 400 | 300 | 180 |

若实际工作电压 U 比额定电压 $U_R$ 低，电容器可工作在更高的 dV/dt 场合。这样 dV/dt 允许值应为右表值乘 $U_R/U$。
If the working voltage (U) is lower than the rated voltage ($U_R$), the capacitor can be worked at a higher dV/dt. In this case, the maximum allowed dV/dt is obtained by multiplying the right value with $U_R/U$.

## 外形尺寸 Dimensions

### 外形尺寸 250VDC Dimensions (mm) 250VDC

| $C_R$ (μF) | W (±1) | H (±1) | T (±1) | P (±1) | DΦ (±0.05) |
|---|---|---|---|---|---|
| **0.01** | 13 | 9 | 4 | 10 | 0.6 |
| **0.015** | 13 | 9 | 4 | 10 | 0.6 |
| **0.022** | 13 | 9 | 4 | 10 | 0.6 |
| **0.033** | 13 | 9 | 4 | 10 | 0.6 |
| **0.047** | 13 | 9 | 4 | 10 | 0.6 |
| **0.068** | 13 | 9 | 4 | 10 | 0.6 |
| **0.1** | 13 | 9 | 4 | 10 | 0.6 |
| **0.15** | 13 | 11 | 5 | 10 | 0.6 |
| **0.22** | 13 | 11 | 5 | 10 | 0.6 |
| **0.33** | 13 | 11 | 5 | 10 | 0.6 |
| **0.47** | 13 | 12 | 6 | 10 | 0.6 |
| **0.68** | 13 | 13 | 7 | 10 | 0.6 |
| **1.0** | 13 | 16 | 8 | 10 | 0.6 |
| **0.33** | 18 | 11 | 5 | 15 | 0.8 |
| **0.47** | 18 | 11 | 5 | 15 | 0.8 |
| **0.68** | 18 | 11 | 5 | 15 | 0.8 |
| **1.0** | 18 | 13.5 | 7.5 | 15 | 0.8 |
| **1.5** | 18 | 14.5 | 8.5 | 15 | 0.8 |
| **2.2** | 18 | 16 | 10 | 15 | 0.8 |

### 外形尺寸 450VDC Dimensions (mm) 450VDC

| $C_R$ (μF) | W (±1) | H (±1) | T (±1) | P (±1) | DΦ (±0.05) |
|---|---|---|---|---|---|
| **0.01** | 13 | 9 | 4 | 10 | 0.6 |
| **0.015** | 13 | 9 | 4 | 10 | 0.6 |
| **0.022** | 13 | 11 | 5 | 10 | 0.6 |
| **0.033** | 13 | 11 | 5 | 10 | 0.6 |
| **0.047** | 13 | 11 | 5 | 10 | 0.6 |
| **0.068** | 13 | 11 | 5 | 10 | 0.6 |
| **0.1** | 13 | 11 | 5 | 10 | 0.6 |
| **0.15** | 13 | 11 | 5 | 10 | 0.6 |
| **0.22** | 13 | 12 | 6 | 10 | 0.6 |
| **0.33** | 13 | 13 | 7 | 10 | 0.6 |
| **0.47** | 13 | 16 | 8 | 10 | 0.6 |
| **0.68** | 13 | 19 | 9 | 10 | 0.6 |

| $C_R$ (μF) | W (±1) | H (±1) | T (±1) | P (±1) | DΦ (±0.05) |
|---|---|---|---|---|---|
| **1.0** | 13 | 21 | 11.5 | 10 | 0.6 |
| **0.33** | 18 | 11 | 5 | 15 | 0.6 |
| **0.47** | 18 | 12 | 6 | 15 | 0.6 |
| **0.68** | 18 | 13.5 | 7.5 | 15 | 0.8 |
| **1.0** | 17.5 | 17.5 | 7.5 | 15 | 0.8 |
| **1.5** | 26.5 | 17 | 8.5 | 22.5 | 0.8 |
| **2.2** | 26.5 | 19 | 10 | 22.5 | 0.8 |

## 外形尺寸 550VDC Dimensions (mm) 550VDC

| $C_R$ (μF) | W (±1) | H (±1) | T (±1) | P (±1) | DΦ (±0.05) |
|---|---|---|---|---|---|
| **0.01** | 13 | 9 | 4 | 10 | 0.6 |
| **0.015** | 13 | 9 | 4 | 10 | 0.6 |
| **0.022** | 13 | 11 | 5 | 10 | 0.6 |
| **0.033** | 13 | 11 | 5 | 10 | 0.6 |
| **0.047** | 13 | 11 | 5 | 10 | 0.6 |
| **0.068** | 13 | 11 | 5 | 10 | 0.6 |
| **0.1** | 13 | 11 | 5 | 10 | 0.6 |
| **0.15** | 13 | 12 | 6 | 10 | 0.6 |
| **0.22** | 13 | 13 | 7 | 10 | 0.6 |
| **0.33** | 18 | 13 | 6.5 | 15 | 0.8 |
| **0.47** | 18 | 14.5 | 8.5 | 15 | 0.8 |
| **0.68** | 18 | 16 | 10 | 15 | 0.8 |
| **1.0** | 18 | 18.5 | 11 | 15 | 0.8 |
| **1.0** | 26.5 | 17 | 8.5 | 22.5 | 0.8 |
| **1.5** | 26 | 20 | 11 | 22.5 | 0.8 |
| **2.2** | 26 | 25 | 16.5 | 22.5 | 0.8 |

## 外形尺寸 630VDC Dimensions (mm) 630VDC

| $C_R$ (μF) | W (±1) | H (±1) | T (±1) | P (±1) | DΦ (±0.05) |
|---|---|---|---|---|---|
| **0.01** | 13 | 9 | 4 | 10 | 0.6 |
| **0.015** | 13 | 9 | 4 | 10 | 0.6 |
| **0.022** | 13 | 11 | 5 | 10 | 0.6 |
| **0.033** | 13 | 11 | 5 | 10 | 0.6 |
| **0.047** | 13 | 11 | 5 | 10 | 0.6 |
| **0.068** | 13 | 11 | 5 | 10 | 0.6 |
| **0.1** | 13 | 12 | 6 | 10 | 0.6 |
| **0.15** | 13 | 13 | 7 | 10 | 0.6 |
| **0.22** | 13 | 16 | 8 | 10 | 0.6 |
| **0.33** | 18 | 14.5 | 8.5 | 15 | 0.8 |
| **0.47** | 18 | 16 | 10 | 15 | 0.8 |
| **0.68** | 18 | 18.5 | 11 | 15 | 0.8 |
| **1.0** | 18 | 22 | 13 | 15 | 0.8 |
| **1.0** | 26 | 20 | 11 | 22.5 | 0.8 |
| **1.5** | 26 | 21.5 | 12 | 22.5 | 0.8 |
| **2.2** | 26 | 25 | 16.5 | 22.5 | 0.8 |

## 备注 Notes

- **产品标识示例 Marking example:** BKW MPB / 104J630V